
PubMedQA v1: Local Evidence-Aware Biomedical Question Answering over PubMed

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Abstract

This report updates the PubMedQA local baseline after the v1 retrieval, answer-generation, and release-gate work. The system combines MedCPT dense retrieval, SQLite FTS5 lexical retrieval, reciprocal-rank fusion, bounded evidence/citation/recency scoring, duplicate suppression, clinical-query-aware context balancing, local MLX generation, second-pass evidence summaries, and PMID citation validation. On a 13-question expanded benchmark, the current citation+recency retrieval baseline returns 130 papers with zero Qwen-judged irrelevant papers, 129/130 papers judged relevance ≥ 2 , and mean judged relevance 2.577/3. The regenerated answer benchmark includes an explicit Evidence basis section in all 13 answers and has zero invalid PMID citations or citation-extraction mismatches. We define v1 release gates covering build integrity, retrieval quality, answer quality, latency, and manual edge-case review.

1. Purpose

The project goal is to answer biomedical questions using local PubMed evidence rather than web search or remote retrieval. Here *PubMedQA* refers to this local project, not an evaluation on the canonical PubMedQA dataset. A local snapshot stores PubMed metadata in SQLite with FTS5 and MedCPT article vectors in LanceDB. Query-time execution is designed to be offline except for optional enrichment and evaluation-time PubMed fetching. This document records the current v1-oriented baseline and supersedes the earlier five-question v4 report.

2. System

Given a question, the system embeds the query with the MedCPT query encoder (Jin et al., 2023), retrieves can-

didates from a LanceDB vector table (LanceDB, 2024), retrieves lexical candidates from SQLite FTS5, and fuses dense and lexical ranks with reciprocal rank fusion (Cormack et al., 2009). Retrieved records are then rescored by a bounded additive score:

$$s_{final} = s_{rel} + \alpha I + B_{overlap} + B_{focus} + B_{evidence}, \quad (1)$$

where s_{rel} is normalized fused relevance, I is a bounded impact term derived from RCR, log citation count, citation rate, and recency when available (Hutchins et al., 2016), and the remaining terms reward query overlap, LLM-extracted focus-keyword overlap, and evidence type. The final context is deduplicated by title similarity and balanced between clinical/review and mechanistic/other papers.

Generation uses a local MLX language model (Apple Machine Learning Research, 2023). Answers are prompted to cite every factual claim with retrieved PMIDs, avoid causal language for observational/mechanistic evidence, distinguish endpoint classes, and include a concise Evidence basis section. A citation validator scans the generated answer and reports any PMID not present in the retrieved context.

The CLI now also supports optional second-pass evidence synthesis. Users can summarize either all retrieved papers or only cited papers, use direct or map-reduce summarization, and export per-paper evidence notes for debugging and factual review.

3. Recent Changes

The current baseline includes five major changes since the earlier v4 paper.

Citation and recency scoring. Retrieval impact scoring now combines RCR/log citation count, citation velocity, and recency. This keeps highly cited established papers competitive while allowing recent papers to surface when citation velocity or recency is informative.

Clinical context balancing. Clinical/outcome-oriented questions prefer clinical/review evidence when enough candidates exist. The usual clinical query target is seven clini-

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cal/review papers and three mechanistic/other papers; mechanistic questions can still return fully mechanistic contexts.

Answer prompt calibration. The answer prompt now explicitly asks for an Evidence basis section and cautions against unqualified “yes” answers for observational associations. It instructs the model to distinguish incidence/prevention, survival after diagnosis, treatment response, surrogate outcomes, and mechanistic findings.

Answer comparison tooling. A new `scripts/compare-answers.py` script compares answer artifacts, including latency, citation validity, citation-set changes, Evidence basis coverage, citation-status mismatches, and simple language review flags.

Build integrity and release gates. Snapshot builds now report duplicate PMIDs, LanceDB row counts, SQLite row counts, abstract/year/citation coverage, and integrity checks. The project now has explicit v1 release gates for build integrity, retrieval quality, answer quality, latency, and manual review.

4. Benchmark

We use the expanded 13-question benchmark in `benchmarks/expanded_questions.txt`. It covers drug safety/effectiveness, prevention, supplementation, cardiovascular and kidney outcomes, lifestyle interventions, mechanistic biology, and prognosis.

The current key outputs are the citation+recency retrieval baseline, the strict Qwen judged retrieval baseline, the Evidence basis answer baseline, the citation/recency review note, and the v1 release-gate checklist. Aggregate comparison reports and artifact filenames are committed in the repository; raw JSON benchmark outputs are generated artifacts and may be regenerated with the commands below.

The retrieval judge is `mlx-community/Qwen3-4B-4bit`, run locally through MLX with a strict JSON rubric. The answer generator is the configured local MLX model.

5. Retrieval Results

Table 1 shows that the current retrieval baseline has no judged irrelevant papers in the top ten across the benchmark. Average mean judge relevance is 2.577/3. Compared with the previous entity-fix baseline, citation+recency scoring changed top-10 membership for only 4/13 questions, preserved an average PMID-set Jaccard overlap of 0.909, increased total Direct@10 papers from 75 to 76, and kept Rel@10 at 129/130. The principal cost is latency: average retrieval time increased from 11.26s to 13.22s in the judged

runs.

6. Answer Generation Results

The regenerated answer benchmark fixed the main prompt-structure gap. Relative to the previous prompt-fix artifact, Evidence basis coverage improved from 0/13 to 13/13. Citation validation remained clean: zero citation warnings, zero invalid cited-PMID rows, and zero manual citation extraction mismatches. Simple review flags in `compare-answers.py` dropped from 13 rows to 0 rows because the missing Evidence basis issue was resolved. Retrieval timings in Table 2 come from a separate end-to-end answer-generation run and therefore differ from the retrieval-only/judged run in Table 1.

7. Known Limitations

First, the local judge is useful for repeatable regression checks but is not a substitute for expert biomedical review. It can be generous for reviews, editorials, and indirect evidence. Second, some clinical questions still retrieve endpoint-mismatched papers, such as survival-after-diagnosis papers for prevention/incidence questions. Third, population constraints require more tuning: for example, adult epilepsy queries can still surface pediatric ketogenic diet papers. Fourth, citation and recency features are useful but not sufficient; stale mechanistic papers can still enter clinical contexts when lexical/vector similarity is high. Fifth, the system retrieves abstracts and metadata, not full text.

8. Reproduction Commands

The current artifacts were produced with the following command pattern, using shortened shell variables to keep line lengths readable:

```
export PUBMEDQA_DATA="/path/to/pubmed_qa"
Q=benchmarks/expanded_questions.txt
RET=retrieval.json
JUD=judged_retrieval.json
ANS=answers.json
OLD=previous_answers.json

./venv/bin/python scripts/evaluate_retrieval.py \
  --questions $Q --out $RET --print-titles

./venv/bin/python scripts/judge_retrieval.py \
  --input $RET --out $JUD --provider mlx \
  --model mlx-community/Qwen3-4B-4bit

./venv/bin/python scripts/evaluate_retrieval.py \
  --questions $Q --with-llm --out $ANS

python3 scripts/compare_answers.py $OLD $ANS
```

9. v1 Release Gates

The v1 release gates require: build-integrity checks to pass or have documented non-blocking warnings; 13/13 benchmark questions returning ten results; no judged irrelevant

Table 1. Current citation+recency retrieval baseline on the expanded benchmark. Direct@10 is the number of papers Qwen judged relevance 3/3. Rel@10 is the number judged at least 2/3.

Question	Retrieval (s)	Clinical	Mech.	Direct@10	Rel@10
Metformin / cancer risk	14.56	7	3	2	10
Statins / dementia	6.37	7	3	8	10
Vitamin D / respiratory infections	12.80	7	3	4	10
Aspirin / primary CVD prevention	20.54	7	3	7	10
GLP-1 / cardiovascular outcomes	13.79	7	3	10	10
Beta blockers / heart failure mortality	13.75	7	3	9	10
SGLT2 inhibitors / kidney outcomes	13.24	7	3	8	9
Omega-3 / cardiovascular prevention	13.29	7	3	6	10
Exercise / insulin sensitivity	13.64	5	5	3	10
Ketogenic diet / adult epilepsy	11.56	5	5	3	10
Aspirin / colorectal cancer risk	10.61	7	3	6	10
Metformin / AMPK mechanism	14.80	0	10	2	10
p53 mutation / cancer prognosis	12.98	2	8	8	10
Mean / aggregate	13.22	5.8	4.2	76/130	129/130

top-10 papers or a documented exception; at least 125/130 papers judged relevance ≥ 2 ; average mean judge relevance at least 2.50; 13/13 answers with an Evidence basis section; zero invalid citations; average retrieval latency below 20s; average answer generation latency below 15s; and manual review of known edge cases including metformin/cancer, aspirin/CRC, aspirin primary prevention, exercise/insulin sensitivity, and p53/prognosis.

The current baseline passes the measured retrieval, answer, and latency gates. The updated build-integrity report still needs to be validated on a fresh snapshot build before all v1 gates can be considered complete. Remaining work before tagging v1 is primarily manual review plus targeted fixes for endpoint, population, and stale-mechanistic retrieval issues.

10. Next Steps

Near-term work should be targeted rather than broad enrichment: verify the improved build-integrity report on a fresh small snapshot; add a stale-mechanistic penalty for clinical outcome questions; improve population/age filtering; strengthen endpoint matching for prevention versus post-diagnosis survival; and use the v1 release gates as the decision checklist for freezing the first release.

References

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Table 2. Current answer baseline with the Evidence basis prompt. Citation validation checks whether generated PMIDs are in the retrieved context.

Question	Retrieval (s)	Generation (s)	Cited PMIDs	Invalid citations
Metformin / cancer risk	15.00	10.05	4	0
Statins / dementia	6.89	8.89	3	0
Vitamin D / respiratory infections	14.84	8.80	5	0
Aspirin / primary CVD prevention	23.70	6.61	5	0
GLP-1 / cardiovascular outcomes	16.17	9.84	4	0
Beta blockers / heart failure mortality	15.89	8.22	3	0
SGLT2 inhibitors / kidney outcomes	14.73	9.65	4	0
Omega-3 / cardiovascular prevention	13.81	9.22	3	0
Exercise / insulin sensitivity	12.83	10.77	5	0
Ketogenic diet / adult epilepsy	10.90	7.99	4	0
Aspirin / colorectal cancer risk	12.25	11.29	5	0
Metformin / AMPK mechanism	15.36	10.71	4	0
p53 mutation / cancer prognosis	14.08	11.14	4	0
Mean	14.34	9.47	4.1	0